

ABTAHI ISLAM

AI & Data Science Enthusiast | Computer Vision & NLP Practitioner

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SUMMARY

Computer Science undergraduate at American International University - Bangladesh with a strong foundation in Artificial Intelligence, Computer Vision, and Natural Language Processing. Experienced in building production-grade, end-to-end ML pipelines spanning medical image segmentation, explainable predictive modelling, and NLP – using Keras, TensorFlow, XG-Boost, LightGBM, and DistilBERT – with rigorous evaluation through SHAP-based interpretability and data-leakage auditing. Published researcher with a peer-reviewed article in IJECE (Q3) and an international conference contribution, with a track record of delivering high-accuracy, deployment-ready AI systems.

TECHNICAL SKILLS

Computer Vision:	CNN, image classification, medical image segmentation, transfer learning (Keras, TensorFlow)
Data Annotation:	Dataset labeling, class balancing, data cleaning, augmentation, leakage auditing
NLP:	Text preprocessing, n-gram language modeling
Data Analysis:	Statistical testing (chi-square, regression), survey design, SPSS, R
Tools:	Python, Optuna, Alumentations, Pandas

WORK EXPERIENCE

AUG. 2024 – PRESENT PRIVATE TUTOR - MATHEMATICS & PROGRAMMING

AI & MACHINE LEARNING PROJECTS

JUL. 2026 – PRESENT

AI-Assisted Early Detection of Alzheimer's Disease from Brain MRI

[🔗 Code](#)

- Built an explainable ML pipeline on 4,333 survey responses to predict 6 mental health outcomes. Optimized XGBoost with Optuna, achieving 75.6% test accuracy and 0.822 ROC-AUC, while integrating SHAP explainability and identifying a data leakage issue that had falsely inflated accuracy.

FEB. 2026 – MAR. 2026

A Comparative Study of Keras Models for Brain Tumor MRI Segmentation

[🔗 Code](#)

- Benchmarked 14 pretrained deep learning models for brain tumor MRI classification using a unified training pipeline, evaluating accuracy, loss, and training time to identify the best performance efficiency trade-offs.

MAR. 2026 – APR. 2026

Trigram Language Model with Story Generation

[🔗 Code](#)

- Built from scratch in Python with Laplace smoothing, trained on 30,000 Shakespeare tokens. Implemented linear interpolation, greedy text generation, and perplexity-based evaluation without using n-gram libraries.

APR. 2026

Automatic Excel Sheet Generator from DOCX Book Data

[🔗 Code](#)

- Automated extraction of chapters, sub-sections, and hadith entries from Bangla .docx files using Python and python-docx. Built a pattern-based pipeline to generate structured Excel datasets while preserving text integrity and eliminating manual data entry.

MAR. 2026 – PRESENT

Explainable Hybrid ML Framework for Post-Breakup Psychological Distress & Suicide Risk

[🔗 Code](#)

- Built a leakage-audited ML pipeline using scikit-learn, XGBoost, LightGBM, and CatBoost on 4,333 survey responses. Achieved 0.793 ROC-AUC for binary high-risk screening and developed a stacking ensemble for multi-class suicide risk prediction.

JUN. 2026

AgriBot – Agricultural Advisory Chatbot for Farmers

[Code](#) · [Live Demo](#)

- Built a 4-layer hybrid AI chatbot using DistilBERT, Random Forest, rule-based reasoning, and a weather API. Developed ML models achieving up to 99.55% accuracy for crop prediction, 98.43% for plant disease detection, and 100% validation accuracy for DistilBERT intent classification with 2× faster inference than BERT.

MAY. 2026

Global Population Growth and Pressure Analysis in R

[Code](#)

- Scraped Worldometers data; built ggplot2 visualisations revealing regional growth trends and policy insights.

PUBLICATIONS

APR. 2026

The ethics of AI technology in academic work: assessing the line between assistance and plagiarism

International Journal of Electrical and Computer Engineering (IJECE) (Q3) · [View Paper](#)

- Reviewed 24 FL oncology studies; compared model architectures and aggregation strategies across institutions.
- Identified privacy-utility trade-offs and proposed practical deployment guidelines for clinical settings.

MAY. 2025

Artificial Intelligence in Textile Waste Management: Exploring Predictive Modelling, Recycling Innovations, and Sustainable Solutions

3rd International Conference on Textile Science and Engineering At: BUTEX · [View Paper](#)

- Proposed a hybrid pipeline for risk detection; achieved strong recall with SHAP-based clinical interpretability.

EDUCATION HISTORY

JAN. 2023 – PRESENT

BACHELOR'S DEGREE, COMPUTER SCIENCE AND ENGINEERING

- American International University-Bangladesh

JAN. 2019 – NOV. 2021

HIGHER SECONDARY CERTIFICATE

- Ishwardi Government College | GPA: 5.00

CERTIFICATION

APR. 2026

Certificate Program in Applied AI (Statistics to NLP) — HCL GUVI

FEB. 2026

Agile Project Management — HP LIFE

JUN. 2023

PC HARDWARE AND SOFTWARE — CISCO

FEB. 2026

Critical Thinking in the AI Era — HP LIFE

NOV. 2025

Gemini Certified University Student — GOOGLE